

TecPal: Making home appliances smarter with rich IoT features

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ABOUT TECPAL

TecPal uses Google Cloud to provide backend services to over a million smart homes across the globe. We make homes smarter, and improve the features of original design manufacturers' products.



About TecPal

Founded in 2018, TecPal is a Hong Kong-based IoT solutions provider that unites cloud technologies, appliances, and lifestyle products for smart homes. Its product designers, engineers, and software specialists help original design manufacturers create an ever-expanding range of smart appliances for homes of the future.

Industries: Technology

Google Cloud results

- 1 second latency for appliance control in Europe and America
- 1 second monitoring latency for over a million appliances in the field
- Delivers required computing power to match increased daily demands
- Scales up service delivery during peak periods such as the holiday season
- Ensures data privacy

Maintain back-end availability over a device

Location: Hong Kong

(<https://cloud.google.com/storage/>), [Compute](#)



About CloudMile

CloudMile is an AI and cloud service provider that focuses on implementing AI to drive business growth. Its mission is to modernize and integrate IT with digital solutions at scale, and it is a Google Cloud partner.

Google Cloud (<https://cloud.google.com/>)

App Engine (<https://cloud.google.com/appengine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

CloudMile is an AI and cloud service

Compute Engine

(<https://cloud.google.com/compute/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud IoT Core (<https://cloud.google.com/iot-core/>)

compliance

across

diverse

jurisdictions,

such as

GDPR in

Europe

Looker Studio

(<https://cloud.google.com/looker-studio>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

When we need help while cooking, many rely on a friend, a family member, or perhaps a trusted cookbook. But imagine if we could tap into thousands of professionally curated recipes to guide us step-by-step at the touch of a button, as we create a tasty dish.

That is what TecPal (<https://www.tecpal.com/>), an Internet of Things (IoT) application development company based in Hong Kong, is achieving with some of the world's leading appliance manufacturers. It develops device and backend software for smart home appliances, recipe databases, management platforms, and companion apps for end-users' smartphones.

"We want to make appliances smart by following a holistic approach to software development. It needs to go hand in hand with hardware design and user experience to deliver the best possible product to the consumer," says Nico Schuster, Chief Executive Officer of TecPal.

"Google Cloud provides more sophisticated solutions to help meet security, privacy, and geographic challenges due to the nature of our worldwide business."

*—Harold Lo, Chief
Technology Officer,
TecPal*

Serving global customers with sophisticated solutions and support

Since its founding in 2018, TecPal has grown to serve over one million devices in customers' homes, with its entire backend service delivery enabled by Google Cloud (<https://cloud.google.com>). "Google Cloud provides more sophisticated solutions to help meet security, privacy, and geographic challenges due to the nature of our worldwide business,"

explains Harold Lo, Chief Technical Officer of TecPal.

Schuster adds that Google's distributed data centers make it easy to address its European clients' GDPR data privacy concerns, for example, while facilitating the working experience for teams in Hong Kong, Shenzhen, Taiwan, Canada, and the United Kingdom.

TecPal developers and engineers have relied on Google Cloud's IoT Core (<https://cloud.google.com/iot-core>) to build its IoT solutions portfolio. However, as IoT Core service is in the process of being migrated to Google-certified partners with support from Google, TecPal has been able to rely on local support teams to make this a smooth transition.

"We've been working very closely with the Google Cloud team to transition our IoT Core development while optimizing connectivity and latency, to improve the appliance control and monitoring experience," says Lo. "Control latency has dropped to as low as 1 second in Europe and America."

Control latency denotes the time when a user inputs a command in the companion app and when it registers in the appliance. Monitoring latency on the other hand, denotes the time it takes for TecPal to register the data that devices in the field are sending. It is in the process of being optimized to achieve a latency of 0.5 seconds.

These optimization efforts have been bolstered with the managed service provider (MSP) from CloudMile

(<https://mile.cloud/>), which has helped TecPal realize cost savings and access to P1 support for its developers and data scientists in Hong Kong and Taiwan, respectively.

"We are still in startup mode and want to grow organically. We're not aiming for hypergrowth, but we must be careful with our business decisions. CloudMile offers us a long-term opportunity to grow with the support level that only companies with revenues with much higher figures would get," Schuster reveals.

"We leverage
Pub/Sub to
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so we won't
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sales event
when we expect
users to perform

over-the-air
updates for their
devices en
masse."

*—Harold Lo, Chief
Technology Officer,
TecPal*

Getting the right amount of computing power to serve millions of people

With the deluge of recipes available online, TecPal needed to ensure that it had enough computing power. "We use [Compute Engine](https://cloud.google.com/compute) (<https://cloud.google.com/compute>) for batch processing as we have recipes in ten languages. These recipes contain a lot of recipe instruction data, and we have several APIs to manage all the recipes displayed on millions of users' mobile apps or appliances," Lo says.

The appliance can customize these recipes for novice and advanced cooks. Experienced cooks can be instructed to cook a dish from scratch, while less experienced cooks can rely on video, images, and text instructions. These recipes also list the right ingredients and nutritional values for each dish.

In the most advanced assisted mode, appliances can run pre-configured device settings, like the right

blade speeds, to cut vegetables for a given recipe. These appliances also always know the best temperature for a given sauce.

"Eventually, we envision using Google Cloud to develop computer vision capabilities that can identify the ingredients in the appliance, like fish or chicken, and then use artificial intelligence to optimize cooking temperatures and timings," Lo explains.

TecPal also uses Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) to automatically scale microservices to serve peak periods like Christmas and Thanksgiving, or even the daily cycle of breakfast, lunch, and dinner.

The company can also scale down the number of pods needed when its architecture realizes that appliances are in standby mode, while obtaining other metrics on the Google Cloud console to estimate future demand and peak times. Ultimately, this helps TecPal reduce computing costs and reserve resources for peak periods.

"On top of that, we also leverage Pub/Sub (<https://cloud.google.com/pubsub>) to manage service event subscriptions to notify us when there's an incoming peak so we won't crash. This can be during a post-sales event when we expect users to perform over-the-air updates for their devices en masse, so these event subscriptions tell us to provide enough computing power," adds Lo.

"Google Cloud offers us the peace of mind to expand what we can offer to our clients . We look forward to discussing how we can implement other Google Cloud services with CloudMile, and the support assurances help us sleep better at night."

*—Nico Schuster, Chief
Executive Officer, TecPal*

Using data to power smarter homes

Currently, TecPal is working towards a new set of tools turning aggregated user data into business

intelligence. Data like appliance errors, appliance performance, and how the user utilizes the appliance are being analyzed by BigQuery (<https://cloud.google.com/bigquery>) and visualized using Looker Studio (<https://cloud.google.com/looker-studio>).

"We are looking at different angles to get the most out of the data not only for us, but also for our clients and ultimately for the end user to help them use the device in a better way. We want to learn how a user might get stuck and then apply a certain logic to the appliance in the background to improve the user experience," says Schuster.

TecPal's future plans encompass growing its hardware development capabilities to include printed circuit board designs to own more hardware development, and augment its IoT connectivity to turn appliances into hubs that can connect with health and wellness services.

"Google Cloud offers us the peace of mind to expand what we can offer to original design manufacturers. We look forward to discussing how we can implement other Google Cloud services with CloudMile, and the support assurances help us sleep better at night," concludes Schuster.